

PROMPT USED FOR “LOG-CONCAVITY OF LEVEL HILBERT FUNCTIONS OF CODIMENSION 3 AND TYPE 2”

Abstract. This document contains the full prompt adapted from the Cycle Double Cover prompt, retargeted at Zanello’s open $(3, 2)$ log-concavity problem for level Hilbert functions.

1 Prompt

Current task statement

Work throughout over an infinite field k (no characteristic restriction unless a proof requires one, in which case the characteristic dependence must be stated explicitly and proved). Let $R = k[x_1, x_2, x_3]$ be a standard graded polynomial ring. An artinian graded algebra is $A = R/I$ for a homogeneous ideal $I \subset R$ containing no nonzero linear forms, with Hilbert function eventually zero. Identify the Hilbert function of A with its h -vector

$$h(A) = (h_0, h_1, \dots, h_e) = (1, 3, h_2, \dots, h_e),$$

where e is the socle degree (the largest index with $h_e > 0$) and $h_i = \dim_k A_i$.

The socle of A is $\text{Ann}_A(\mathfrak{m}_A)$, where $\mathfrak{m}_A$ is the maximal homogeneous ideal. A is *level of type* t if its socle is concentrated in degree e and has dimension t . Equivalently, via Macaulay inverse systems, level algebras of codimension 3 and type t correspond to R -submodules of $S = k[y_1, y_2, y_3]$ generated by t forms of degree e , with h_i equal to the dimension of the span of the degree- i partial derivatives of those generators.

A finite sequence of nonnegative integers $(a_0, a_1, \dots, a_e)$ is *log-concave* if

$$a_{i-1} a_{i+1} \leq a_i^2$$

for every index i with $1 \leq i \leq e - 1$.

Let $\mathcal{S}_{3,2}$ be the set of all Hilbert functions of artinian level algebras of codimension 3 and type 2 (i.e., all sequences of the form $(1, 3, h_2, \dots, h_e = 2)$ that arise as $h(A)$ for some such A).

Resolve the following conjecture completely:

Every Hilbert function in $\mathcal{S}_{3,2}$ is log-concave.

In other words: every artinian level graded algebra of codimension 3 and Cohen–Macaulay type 2 has a log-concave Hilbert function, in every characteristic.

Assume for purposes of this task that a complete affirmative proof exists. A complete solution must prove exactly the following:

Every $h \in \mathcal{S}_{3,2}$ satisfies $h_{i-1}h_{i+1} \leq h_i^2$ for all relevant i , with no extra assumptions such as monomiality (pure O -sequences), characteristic zero, bounded socle degree, unimodality hypotheses, flawlessness, differentiability of the first half, or the Interval Conjecture.

Partial progress does not count unless it implies exactly the resolution above. In particular, proofs only for pure O -sequences / monomial level algebras, proofs only in characteristic zero, proofs only up to a fixed socle degree, computational verification through any fixed bound, reductions to another unproved conjecture (including unimodality, flawlessness, first-half differentiability, Lefschetz properties, or the Interval Conjecture for $\mathcal{S}_{3,2}$), and candidate counterexamples without a complete nonexistence certificate for the affirmative statement are insufficient.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as “N agents for strategy X.” Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations: Macaulay inverse systems and pairs of forms in three variables; apolar ideals and catalecticant / Hankel rank constraints; combinatorial numerical semigroups and O -sequence / Macaulay bound arguments; induction on socle degree; deformation and specialization from general forms; Gotzmann persistence and Hilbert-scheme / flat-limit arguments; Lefschetz-type operators and multiplication maps by linear forms; generating-function and log-concave polynomial identities; explicit classification of short socle-degree cases; and computational sanity checks on random inverse-system generators.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical idea they are using, not by superficial wording. If many agents converge to one family, redirect some of them toward underexplored formulations.
- Do not allow one approach to dominate merely because it gives elegant reductions. A route that ends at a lemma equivalent in strength to the original conjecture (e.g., “all of $\mathcal{S}_{3,2}$ is unimodal,” “all of $\mathcal{S}_{3,2}$ is flawless,” or “WLP/SLP holds for related Gorenstein algebras”) is not close to completion unless it supplies a genuinely new proof of that lemma *and* derives log-concavity from it.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Only continue assigning agents to it if someone proposes a materially new mechanism, invariant, or construction.
- Keep several incompatible proof routes alive through multiple rounds. Cross-pollinate ideas

only after independent agents have developed them far enough to expose their real strengths and gaps.

- Use adversarial agents throughout: every candidate proof must be checked for off-by-one index errors in the log-concavity inequalities; failure at the socle end ($h_{e-2}h_e \leq h_{e-1}^2$ with $h_e = 2$); confusion between level and Gorenstein; accidental restriction to monomial / pure O -sequences; characteristic-dependent steps stated as characteristic-free; misuse of Macaulay bounds as if they forced log-concavity; and circular appeal to an equivalent open property of $\mathcal{S}_{3,2}$.
- Require agents to return concrete lemmas, constructions, equations, explicit inverse-system generators, or counterexamples to proposed sublemmas. Reject status reports, vague optimism, and claims that an unproved global compatibility statement is “routine.”
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Produce a complete proof if one survives audit; otherwise report only the strongest rigorously proved derivation and its exact remaining gap.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete affirmative proof has been found and survives adversarial audit. Do not return a reduction, partial result, isolated missing lemma, “best effort” summary, or explanation of why the problem is difficult.

Spend at least 8 hours on this before even thinking of returning or giving up.

Public search may be used only for ordinary mathematical background or standard named theorems, not to search for a solution to this exact conjecture or benchmark. Do not search the public web merely to determine whether the $(3, 2)$ log-concavity problem is open, and do not answer that it is open.